Graph Convolutional Subspace Clustering: A Robust Subspace Clustering Framework for Hyperspectral Image

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***Abstract*— Hyperspectral image (HSI) clustering is a challenging task due to the high complexity of HSI data. Subspace clustering has been proven to be powerful for exploiting the intrinsic relationship between data points. Despite the impressive performance in the HSI clustering, traditional subspace clustering methods often ignore the inherent structural information among data. In this article, we revisit the subspace clustering with graph convolution and present a novel subspace clustering framework called graph convolutional subspace clustering (GCSC) for robust HSI clustering. Specifically, the framework recasts the self-expressiveness property of the data into the non-Euclidean domain, which results in a more robust graph embedding dictionary. We show that traditional subspace clustering models are the special forms of our framework with the Euclidean data. On the basis of the framework, we further propose two novel subspace clustering models by using the Frobenius norm, namely efficient GCSC (EGCSC) and efficient kernel GCSC (EKGCSC). Each model has a globally optimal closed-form solution, making it easier to implement, train, and apply in practice. Extensive experiments strongly evidence that EGCSC and EKGCSC dramatically outperform current models on three popular HSI data sets consistently.**

***Index Terms*— Graph convolutional networks (GCNs), hyperspectral image (HSI) clustering, kernel method, subspace clustering.**

I. Introduction

HYPERSPECTRAL images (HSIs) acquired by remote sensors contain rich spectral and spatial information, which enable us to accurately recognize the region of interest. Over the past decade, HSIs have been widely applied in various fields, ranging from geological exploration, marine monitoring, military reconnaissance to medical imaging and forensics [1]–[3].

HSI classification, which aims to classify every pixel with a certain label, is the foundation for the application of HSI [3], [4]. The most commonly used HSI classification method is the supervised classification [5], [6] based on label information. In recent years, the supervised HSI classification has made significant progress, which achieved excellent performance dealing with several popular HSI data sets, including Indian Pines, Salinas, and Pavia University images [1], [7]. Particularly, deep learning models [1], [8]–[10], such as convolutional neural networks (CNNs) [11], [12], have crucially narrowed the gap between human and machines. Unfortunately, the supervised HSI classification methods typically require a large amount of labeled samples, which are not available in HSI scenarios due to the high cost of labeling training data. Furthermore, the supervised methods are difficult to deal with unknown objects, since they are modeled with the existing classes.

To avoid manual data annotation, many works dedicate to developing unsupervised HSI classification methods, namely HSI clustering. Instead of using the label information, the HSI clustering aims to find the intrinsic relationship between data points and automatically determines labels in an unsupervised manner [13]. The key to the HSI clustering is to measure the similarity between data points [14]. Traditional clustering methods, e.g., K-means [15], frequently use the pairwise distance as the similarity measurement, such as the Euclidean distance. Owing to the mixed pixel and the redundant band problem [16], [17], these methods often suffer from unreliable measurement, which makes the HSI clustering greatly challenging. Compared with the supervised classification, there are quite fewer studies on the HSI clustering [18] and they are usually uncompetitive in terms of accuracy.

Recently, subspace clustering [19] has drawn increasing attention in the HSI clustering [20]–[24] due to its ability to handle high-dimensional data and its reliable performance. Technically, the subspace clustering seeks to express the data points as a linear combination of a self-expressive dictionary in the same subspace [25]. The subspace clustering model typically consists of two steps, i.e., self-representation [26] and spectral clustering (SC) [24]. To improve the performance

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of the subspace clustering, many works have devoted to constructing a robust affinity matrix by using various techniques. For example, sparse subspace clustering (SSC) [27] uses an ℓ_1-norm to encourage a sparse affinity matrix, while low rank subspace clustering (LRSC) [28] adopts a nuclear norm to enforce the affinity matrix to be low-rank. By considering the spectral and spatial properties of HSIs, Zhang *et al.* proposed a spectral–spatial SSC (S^4C) [23]. Kernel subspace clustering [29] was proposed as the nonlinear extension of the subspace clustering model by implicitly mapping data into higher kernel space. In [21], an improved kernel subspace clustering was applied to the HSI clustering.

However, the previous subspace clustering models are typically based on the Euclidean data, which results in the neglect of the inherent graph-structured information contained in the HSI data points. Owing to the interference of the imaging spectrometer and the atmosphere, the HSI data points usually get corrupted by noise. The impact can cause entries with large errors as well [30]. As a result, the traditional subspace models are vulnerable to the noisy data and outliers in the Euclidean domain. Although manifold regularization [18], [31] has been proven to be useful to incorporate graph information into the subspace clustering, such as graph regularized LRSC [30], [32], it essentially relies on the pure Euclidean data rather than the structural data. Moreover, manifold regularization often suffers from additional regularization terms and regularization coefficients. In contrast, these drawbacks can be effectively alleviated or even avoided by turning the HSI clustering into a clustering problem in the non-Euclidean domain. The recent development of graph neural networks (GNNs) [33]–[35] generalizes the powerful CNNs in dealing with the Euclidean data to modeling the graph-structured data. This allows us to revisit traditional problems in the non-Euclidean domain. Unfortunately, not enough attention has been paid to developing a robust model for HSI processing in the non-Euclidean domain.

To learn graph embedding and affinity simultaneously, in this article, we present a graph convolutional subspace clustering (GCSC) framework that recasts the traditional subspace clustering into the non-Euclidean domain. Specifically, the GCSC framework calculates the self-representation coefficient matrix by leveraging a graph convolutional self-representation model combining both graph and feature information. As a result, the proposed framework can circumvent noise data and tends to produce a more robust affinity than the traditional subspace clustering models. We provide an efficient implementation of the GCSC framework [i.e., efficient GCSC (EGCSC)] by introducing the Frobenius norm, which is further extended to the nonlinear subspace [i.e., efficient kernel (EKGCSC)] by using the kernel trick. Visually, an intuitive description about the motivation of GCSC is illustrated in Fig. 1.

To sum up, the main contributions of this work are follows.

1) A robust subspace clustering framework, GCSC, is developed for the HSI clustering in which the subspace clustering is recast into the non-Euclidean domain. Particularly, the traditional subspace clustering can be viewed as the special form of the proposed framework.
2) Based on the Frobenius norm, two novel and efficient subspace clustering models are proposed under the GCSC framework. We refer to them as EGCSC and EKGCSC, respectively. The two methods have a closed-form solution, thus making them easier to implement, train, and apply in practice.
3) Our experimental results on several HSI data sets show that the proposed subspace clustering models are effectively better than many existing clustering methods for the HSI clustering. The successful attempt of GCSC offers an alternative orientation for unsupervised learning.

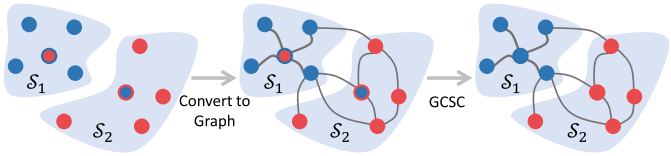


Fig. 1. Motivation of GCSC. In the figure, the blue and red points signify that two different classes that lie in the subspace $\mathcal{S}_1$ and $\mathcal{S}_2$, respectively. The red point with a blue outline and the blue point with a red outline denote the misclassified points. Intuitively, GCSC converts the traditional data into the graph-structured data and adopts the graph convolution to generate the robust embedding for the subsequent subspace clustering.

The rest of this article is structured as follows. We first briefly review the subspace clustering, graph convolutional networks (GCNs), and HSI clustering in Section II. Second, we describe the details of the developed GCSC framework and its two implementations in Section III. In Section IV, we give extensive experimental results and empirical analysis. Finally, we conclude with a summary and future works in Section V.

II. Preliminaries and Related Work

A. Notations

Throughout this article, boldface lowercase italics symbols (e.g., $\boldsymbol{x}$), boldface uppercase roman symbols (e.g., $\mathbf{X}$), regular italics symbols (e.g., x_{ij}), and calligraphy symbols (e.g., $\mathcal{S}$) denote vectors, matrices, scalars, and sets, respectively. A graph is represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ denotes the node set of the graph with $v_i \in \mathcal{V}$ and $|\mathcal{V}| = N$, $\mathcal{E}$ indicates the edge set with $(v_i, v_i) \in \mathcal{E}$, and $\mathbf{A} \in \mathbb{R}^{N\times N}$ stands for an adjacency matrix. We define the diagonal degree matrix of the graph as $\mathbf{D} \in \mathbb{R}^{N\times N}$, where $D_{ii} = \sum_j A_{ij}$. The graph Laplacian is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, and its normalized version is given by $\mathbf{L}_{sym} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$. In this article, $\mathbf{X}^T$ denotes the transpose of matrix $\mathbf{X}$, and $\mathbf{I}_N$ denotes an identity matrix with the size of N. The Frobenius norm of a matrix is defined as $\|\mathbf{X}\|_F = (\sum_{ij} |x_{ij}|^2)^{1/2}$, and the trace of a matrix is denoted as $\mathrm{tr}(\mathbf{X})$.

B. Subspace Clustering Models

Let $\mathbf{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{m\times N}$ be a collection of N data points $\{\boldsymbol{x}_i \in \mathbb{R}^m\}_{i=1}^N$ drawn from a union of linear or affinity subspaces $\mathcal{S}_1 \cup \mathcal{S}_2 \cup \cdots \cup \mathcal{S}_n$, where N, m, and n denote the number of data points, features, and subspaces, respectively. The subspace clustering model for the given data

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set $\mathbf{X}$ is defined as the following self-representation problem [27], [32]:

$$\min\|\mathbf{W}\|_p \quad \text{s.t. } \mathbf{XW} = \mathbf{X}, \quad \text{s.t. } \text{diag}(\mathbf{W}) = 0 \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{N\times N}$ denotes the self-expressive coefficient matrix and $\text{diag}(\mathbf{W}) = 0$ enforces the diagonal elements of $\mathbf{W}$ to be zero so that the trivial solutions are avoided. $\|\mathbf{W}\|_p$ denotes a p-norm of matrix $\mathbf{W}$, e.g., $\|\mathbf{W}\|_1$ (SSC) [27]), and $\|\mathbf{W}\|_2$ (ℓ_2-SSC [22]).

In the SSC model, the self-expressive coefficient matrix is assumed to be sparse and the self-representation problem is often formulated as

$$\arg\min_{\mathbf{W}} \|\mathbf{XW} - \mathbf{X}\|_F^2 + \lambda\|\mathbf{W}\|_1, \quad \text{s.t. } \text{diag}(\mathbf{W}) = 0. \tag{2}$$

Here, the ℓ_1-norm tends to produce a sparse coefficient matrix. By using a nuclear norm, LRSC [32], [36], [37] reformulates the self-expressiveness property of data as

$$\arg\min_{\mathbf{W}} \|\mathbf{XW} - \mathbf{X}\|_{2,1}^2 + \lambda\|\mathbf{W}\|_*, \quad \text{s.t. } \text{diag}(\mathbf{W}) = 0 \tag{3}$$

where $\|\cdot\|_*$ and $\|\cdot\|_{2,1}$ denote the nuclear norm and $\ell_{2,1}$-norm of a matrix. LRSC has been proven to be effective to incorporate the global structure of data. Furthermore, subspace clustering can be used to model corrupted data, i.e., $\mathbf{X} = \mathbf{XW} + \mathbf{N}$, where $\mathbf{N}$ is arbitrary noise.

The above problems can be efficiently solved by using convex optimization methods, such as alternating direction method of multipliers (ADMM) [19], [38]. Once the coefficient matrix $\mathbf{W}$ is found, the subspace clustering seeks to segment an affinity matrix $\mathbf{A} = (1/2)(|\mathbf{W}| + |\mathbf{W}|^T)$ by SC method [24].

C. GCNs

There is an increasing interest in generalizing convolutions to the graph domain [33], [39]. The recent development in GNNs allows us to efficiently approximate convolution on the graph-structured data. GNNs can typically divide into two categories [34], [35]: spectral convolutions, which perform convolution by transforming node representations into the spectral domain using the graph Fourier transform or its extensions, and spatial convolutions, which perform convolution by considering node neighborhoods. Unless otherwise specified, the graph convolution involved in this article is the spectral convolution.

One of the most representative graph convolution models is the GCNs developed by Kipf and Welling [40]. GCN simplifies the spectral convolution by approximating spectral filters with the 1th-order Chebyshev polynomials and setting the largest eigenvalue of the normalized graph Laplacian $\mathbf{L}_{\text{sym}}$ to 2. Formally, GCN defines spectral convolution over a graph as follows:

$$\mathbf{H} = \sigma\left(\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\mathbf{X}^T\mathbf{W}\right). \tag{4}$$

Here, $\tilde{\mathbf{A}} = \mathbf{I}_N + \mathbf{A}$ is an adjacency matrix with self-loops, $\tilde{\mathbf{D}}$ denotes the degree matrix of nodes whose elements are given by $\tilde{D}_{ii} = \sum_j \tilde{A_{ij}}$, $\mathbf{W} \in \mathbb{R}^{m\times l}$ denotes a trainable parameter matrix, and σ is a nonlinear activation function. Specifically, GCN takes a node feature matrix $\mathbf{X}$ and an adjacency matrix $\mathbf{A}$ as inputs, and produces a graph embedding $\mathbf{H} \in \mathbb{R}^{N\times l}$, where l is the output dimension.

GCN is originally developed for semisupervised node classification. By stacking several graph convolution layers, it is possible for GCN to learn deeper graph representation. Similar to traditional deep neural networks, GCN can be easily trained using gradient descent methods. In different tasks, the GCN models allow many traditional problems to be revised in the non-Euclidean domain.

D. HSI Clustering

Although supervised methods have achieved great success in HSI classification [5], [9], they are limited by the lack of sufficient labeled data. To avoid data annotation, HSI clustering has attracted increasing attention. HSI clustering is based on the fact that the same land-cover object often shows similar spectral curves. The traditional centroid-based clustering methods, such as K-means [41] and fuzzy c-means (FCM) [42], are widely used and easy to implement. However, these methods are sensitive to the random initialization state and thus their clustering results are hard to reproduce [18].

With the advantage of its stable performance, subspace clustering is frequently used for HSI clustering. Huang *et al.* [20], Zhai *et al.* [21], [22], and Zhang *et al.* [23] have successfully applied various subspace clustering methods to the HSI clustering, including S^4C [23], kernel SSC [21], and joint sparsity-based SSC (JSSC) [20]. Benefited from the ability to exploit the intrinsic structure of data, subspace clustering has achieved impressive performance. Recently, evolutionary optimization based clustering method has attracted increasing interests, e.g., evolutionary multiobjective optimization-based HSI clustering [43], [44]. It is well known that the evolutionary algorithm is powerful to search the globally optimal solution but it often results in huge computational cost [45]–[48].

It has been proven to be effective to improve HSI clustering performance by utilizing spectral and spatial information, simultaneously. Zhang *et al.* [18] developed state-of-the-art HSI clustering method called robust manifold matrix factorization (RMMF) clustering by combining HSI dimensionality reduction and data clustering, simultaneously. Kong *et al.* [49] proposed an unsupervised broad learning (UBL) clustering method that combines clustering with broad representation learning. In our recent work [50], we proposed a deep subspace clustering method for the HSI clustering, which further demonstrates the potential of combining clustering models with feature learning.

III. Methodology

In this section, we first introduce the proposed GCSC framework. Then, we provide the details of two novel subspace clustering models based on the framework, i.e., EGCSC and EKGCSC. We illustrate a schematic of the proposed framework in Fig. 2 and more details are given in the following sections.

A. GCSC Framework

Inspired by the recent development of GCNs, we present a novel subspace clustering framework by incorporating graph

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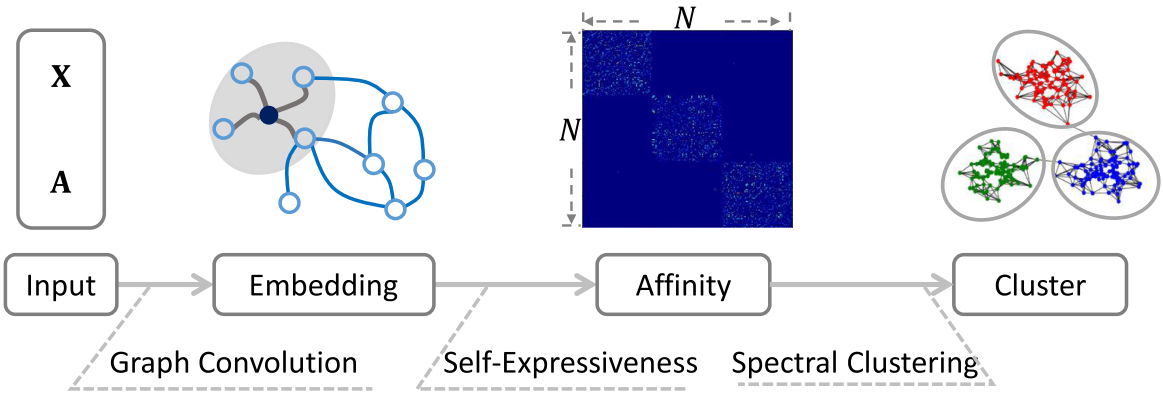


Fig. 2. Flowchart of the proposed GCSC framework. The framework follows the basic steps of the existing subspace clustering methods but takes a feature matrix $\mathbf{X}$ and an adjacency matrix $\mathbf{A}$ of the graph as inputs, which results in a novel graph convolutional self-representation. The graph convolution will generate a robust graph embedding that is used as a dictionary for the subsequent affinity learning.

embedding into subspace clustering. We refer to the framework as GCSC. The goal of the GCSC framework is to utilize graph convolution to learn a robust affinity. For this purpose, we first modify the traditional self-representation as follows:

$$\mathbf{X} = \mathbf{X}\bar{\mathbf{A}}\mathbf{Z}, \quad \text{s.t. } \operatorname{diag}(\mathbf{Z}) = 0. \tag{5}$$

Here, $\mathbf{Z} \in \mathbb{R}^{N\times N}$ is the self-representation coefficient matrix, and $\bar{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$ denotes the normalized adjacency matrix with self-loops. Notably, $\mathbf{X}\bar{\mathbf{A}}\mathbf{Z}$ can be treated as a special linear graph convolution operation (or a special graph autoencoder) parameterized by $\mathbf{Z}$, which embeds a graph into a vector space where the structure and the inherent properties of the graph are preserved. We call (5) graph convolutional self-representation.

Parallel to traditional subspace clustering models, the GCSC framework can be rewritten as

$$\arg\min_{\mathbf{Z}} \frac{1}{2}\left\|\mathbf{X}\bar{\mathbf{A}}\mathbf{Z} - \mathbf{X}\right\|_q + \frac{\lambda}{2}\|\mathbf{Z}\|_p, \quad \text{s.t. } \operatorname{diag}(\mathbf{Z}) = 0 \tag{6}$$

where q and p denote any appropriate matrix norm, such as $p, q = 0, (1/2), 1, 2$ and nuclear norm, and λ is a tradeoff coefficient. It is easy to prove that the traditional subspace models are the special cases of our framework, i.e., the traditional subspace models depend only on data features. For example, when q is the Frobenius norm and $p = 1$, (6) becomes the extension of the classical SSC [27], while q is $\ell_{2,1}$ norm and p is the nuclear norm, (6) degenerates to LRSC [28], [37]. Equation (6) can be effectively solved by the same method adopted in the traditional subspace clustering. Once the self-representation coefficient matrix $\mathbf{Z}$ is obtained, SC can be used to generate the clustering results.

B. EGCSC

On the basis of the proposed framework, we present the first novel subspace clustering model, namely EGCSC, by setting both q and p to be the Frobenius norm. Formally, we formulate the EGCSC model as

$$\arg\min_{\mathbf{Z}} \frac{1}{2}\left\|\mathbf{X}\bar{\mathbf{A}}\mathbf{Z} - \mathbf{X}\right\|_F^2 + \frac{\lambda}{2}\|\mathbf{Z}\|_F^2. \tag{7}$$

Ji *et al.* [51] have proven that the Frobenius norm will not result in trivial solutions even without constraint $\operatorname{diag}(\mathbf{Z}) = 0$. This leads to a dense self-representation coefficient matrix and an efficient closed-form solution. The solution is given by

$$\mathbf{Z} = \left(\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}} + \lambda\mathbf{I}_N\right)^{-1}\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}. \tag{8}$$

The proof of (8) is given in the Appendix. Unlike traditional subspace clustering models that often need alternative optimization, EGCSC guarantees high computational efficiency.

Having obtained $\mathbf{Z}$, we can use it to construct an affinity matrix $\mathbf{C}$ for the SC. However, there is no globally accepted solution for this step in the literature. Most existing works typically compute the affinity matrix by $\mathbf{C} = |\mathbf{Z}| + |\mathbf{Z}|^T$ or $|\mathbf{Z}|$. In this article, we use the heuristic adopted by efficient dense subspace clustering (EDSC) [51] to enhance the block-structure, which is proved beneficial for clustering accuracy. The pseudocode of the EGCSC is given in Algorithm 1.

Algorithm 1 EGCSC

Input: $\mathbf{X}$, $\mathbf{A}$, λ, and the number of clusters.
1 Compute $\bar{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$;
2 Compute coefficient matrix:
$\mathbf{Z} = \left(\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}} + \lambda\mathbf{I}_N\right)^{-1}\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}$;
3 Construct affinity matrix $\mathbf{C}$;
4 Apply spectral clustering on $\mathbf{C}$;
Output: Clustering results.

C. EKGCSC

We have proposed the EGCSC method. However, the EGCSC model is essentially modeled on linear subspaces. Due to the complexity and nonlinearity of HSI, a large number of works have demonstrated that nonlinear models will yield better performance than their linear counterparts. In this section, we provide a nonlinear extension of EGCSC by using the kernel trick. The extension is referred to as EKGCSC.

Let $\Phi : \mathbb{R}^m \to \mathcal{H}$ be a mapping from the input space to the reproducing kernel Hilbert space $\mathcal{H}$. We define a positive

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semidefinite kernel Gram matrix $\mathbf{K_{XX}} \in \mathbb{R}^{N\times N}$ as

$$[\mathbf{K_{XX}}]_{ij} = \left[\langle \Phi(\mathbf{X}), \Phi(\mathbf{X})\rangle_{\mathcal{H}}\right] = \Phi(\boldsymbol{x}_i)^T\Phi(\boldsymbol{x}_j) = \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) \tag{9}$$

where $\kappa : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}$ denotes the kernel function. In this article, the Gaussian kernel is used, i.e., $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp(-\gamma\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2)$, where γ is the parameter of the Gaussian kernel function. Formally, the EKGCSC model is expressed as

$$\arg\min_{\mathbf{Z}} \frac{1}{2}\left\|\Phi(\mathbf{X})\bar{\mathbf{A}}\mathbf{Z} - \Phi(\mathbf{X})\right\|_F^2 + \frac{\lambda}{2}\|\mathbf{Z}\|_F^2. \tag{10}$$

By using kernel trick, (10) can be equivalently rewritten as

$$\arg\min_{\mathbf{Z}} \frac{1}{2}\mathrm{tr}\big(\mathbf{Z}^T\bar{\mathbf{A}}^T\mathbf{K_{XX}}\bar{\mathbf{A}}\mathbf{Z} - 2\mathbf{K_{XX}}\bar{\mathbf{A}}\mathbf{Z} + \mathbf{K_{XX}} + \lambda\mathbf{Z}^T\mathbf{Z}\big). \tag{11}$$

The above problem can be solved by calculating the partial derivative with respect to $\mathbf{Z}$ and set it to be zero (see the Appendix). The closed-form solution of EKGCSC is given by

$$\mathbf{Z} = \left(\bar{\mathbf{A}}^T\mathbf{K_{XX}}\bar{\mathbf{A}} + \lambda\mathbf{I}_N\right)^{-1}\bar{\mathbf{A}}^T\mathbf{K_{XX}}. \tag{12}$$

The EKGCSC model explicitly maps the original data points onto a higher dimensional space, and thus makes a linearly inseparable problem to be a separable one. We use a manner that is similar to EGCSC to construct the affinity matrix and obtain the final clustering results by SC. The pseudocode of EKGCSC is given in Algorithm 2.

Algorithm 2 EKGCSC

Input: $\mathbf{X}$, $\mathbf{A}$, λ, kernel parameters, and the number of clusters.

1 Compute $\bar{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$;
2 Compute kernel matrix $\mathbf{K_{XX}}$ according to Eq. (9);
3 Compute coefficient matrix: $\mathbf{Z} = \left(\bar{\mathbf{A}}^T\mathbf{K_{XX}}\bar{\mathbf{A}} + \lambda\mathbf{I}_N\right)^{-1}\bar{\mathbf{A}}^T\mathbf{K_{XX}}$;
4 Construct affinity matrix $\mathbf{C}$;
5 Apply spectral clustering on $\mathbf{C}$;

Output: Clustering results.

D. HSI Clustering Using the GCSC Models

We use the proposed GCSC models for HSI clustering. Two essential issues need to be tackled before using the GCSC models. First, HSI data often includes many spectral bands with lots of redundancy, and thus using only spectral features is hard to achieve good performance. Second, the GCSC models are based on the graph-structured data, and however, HSI is typically a Euclidean data.

To remedy the first issue, the following procedures are employed. We first use principal component analysis (PCA) to reduce the spectral dimensionality by preserving the top d PCs. On the one hand, PCA reduces the redundant information contained in HSI data. On the other hand, it increases computational efficiency when model training. To take spectral and spatial information into consideration, simultaneously, we represent every data point by extracting 3-D patches. Specifically, every data point is represented by the center pixel and its neighboring pixels. The manner is widely adopted in different HSI spectral–spatial classification methods [2], [7], [9].

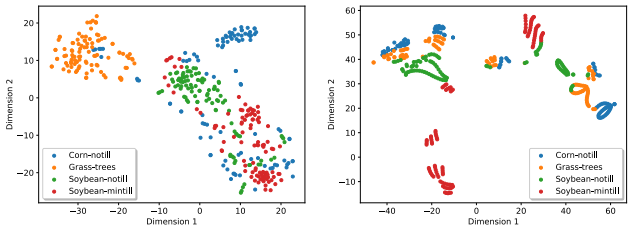


Fig. 3. Visualization of data points selected from Indian Pines data set, where we randomly select 100 data points per class and reduce their feature dimensionality into 2 with t-SNE [52]. (a) Original data points. (b) Embedding using graph convolution. It can be seen that corn-till, soybean-notill, and soybean-mintill are mixed in the original data distribution. By contrast, they show better separability and are more compact after transformed by graph convolution.

For the second issue, we construct a k-nearest neighbor (kNN) graph to represent the graph structure of the data points. Specifically, each data point is viewed as a node over the graph and the kNNs of $\boldsymbol{x}_i$ consists of the edge relationship. The adjacent matrix $\mathbf{A}$ of a kNN graph is defined by

$$A_{ij} = \begin{cases} 1, & \boldsymbol{x}_j \in \mathcal{N}_k(\boldsymbol{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where $\mathcal{N}_k(\boldsymbol{x}_i)$ indicates the kNNs of $\boldsymbol{x}_i$, which is determined by computing the pairwise Euclidean distance between the PCA reduced data points.

E. Remarks on the Proposed GCSC

In this section, we provide a deeper insight into the GCSC framework from the following viewpoints. Let $\mathbf{Y} = \mathbf{X}\bar{\mathbf{A}}$ be the graph embedding, thus GCSC can be rewritten as

$$\arg\min_{\mathbf{Z}} \frac{1}{2}\|\mathbf{YZ} - \mathbf{X}\|_q + \frac{\lambda}{2}\|\mathbf{Z}\|_p, \quad \text{s.t. } \mathrm{diag}(\mathbf{Z}) = 0. \tag{14}$$

From the viewpoint of sparse representation, GCSC aims to use a self-expressive dictionary matrix $\mathbf{Y}$ to reconstruct the original data. Since $\mathbf{Y}$ considers the global structure information, those noise points will be eliminated and a clear dictionary can be obtained, which is beneficial for producing a robust affinity matrix. It can be seen from Fig. 3, the resulting $\mathbf{Y}$ shows better clustering characteristics than the original $\mathbf{X}$. It can be further explained from the viewpoint of graph representation learning. Graph convolution essentially is a special form of the Laplacian smoothing [53], which combines the features of a node and its nearby neighbors. The operation makes the features of the node in the same cluster similar, thus greatly easing the clustering task.

The main differences between the GCSC model and the traditional subspace model are as follows. First, GCSC is built in the non-Euclidean domain. Under the GCSC framework, the traditional subspace clustering models can be considered as special cases in the Euclidean domain. Second, GCSC

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TABLE I
SUMMARY OF SALINAS, INDIAN PINES, AND PAVIA UNIVERSITY DATA SETS

Datasets	SalinasA	Indian Pines	Pavia University
Pixels	83×86	85×70	140×150
Channels	204	200	103
Clusters	6	4	8
Samples	5348	4391	6445
Sensor	AVIRIS	AVIRIS	ROSIS

TABLE II
SETTINGS OF THE IMPORTANT HYPERPARAMETERS IN EGCSC AND EKGCSC

Method	EGCSC		EKGCSC		
	λ	k	λ	k	γ
SalinasA	100	30	100	30	0.2
Indian Pines	100	30	100000	30	6
Pavia University	1000	20	60000	30	100

incorporates the graph structure via graph convolution, while the traditional subspace clustering models do this by manifold regularization. Therefore, GCSC exploits graph information using a more straightforward way.

IV. EXPERIMENTS

In this section, we extensively evaluate the clustering performance of the proposed clustering methods on three frequently used HSI data sets. The source code of EGCSC and EKGCEC are released at https://github.com/AngryCai/GraphConvSC.

A. Setup

1) Data Sets and Preprocessing: We conduct experiments on three real HSI images acquired by AVIRIS and ROSIS sensors, i.e., Salinas, Indian Pines, and Pavia University. Most subspace clustering models require the calculation of the pairwise relationship between data points one time, and thus, they often suffer from an out of memory problem during the training. For computational efficiency, we separately take a subscene of these data sets for evaluation as it is done in [21], [23], [49], [50], and [54]. Specifically, these subscenes are located within the original scenes at $[591-676, 158-240]$, $[30-115, 24-94]$, and $[150-350, 100-200]$, respectively. Notice that the subscene taken from the Salinas data set is also known as the SalinasA data set. The details of the three data sets are summarized in Table I.

In data preprocessing, we perform PCA to reduce spectral bands into 4 by preserving at least 95% of the cumulative percentage of variance (CPV). We construct spectral–spatial samples by setting neighborhood size to be 9 for all the data sets. All data points are standardized by scaling into $[0, 1]$ before clustering.

2) Evaluation Metrics: Three popular metrics [18], [23], [49] are used to evaluate the clustering performance of clustering models, i.e., overall accuracy (OA), normalized mutual information (NMI), and kappa coefficient (Kappa). These metrics range in $[0, 1]$, where the higher the scores are, the more accurate the clustering results are achieved. Besides, to evaluate the computational complexity of our models, running time of different methods is compared in the experiment.

3) Compared Methods: We compare the proposed methods with several existing HSI clustering methods, including traditional clustering methods and state-of-the-art methods. Specifically, the compared traditional clustering methods contain SC [24], SSC [27], EDSC [51], LRSC [36], and ℓ_2-norm based SSC (ℓ_2-SSC) [22]. The compared state-of-the-art HSI clustering methods are S^4C [23], UBL clustering [49], and RMMF [18].

For these HSI clustering methods, i.e., ℓ_2-SSC, S^4C, UBL, and RMMF, we follow their settings reported in the corresponding literature. The hyperparameters of EGCSC and EKGCSC are given in Table II. All the compared methods are implemented with Python 3.5 running on an Intel Xeon E5-2620 2.10-GHz CPU with 32-GB RAM.

B. Results

1) Quantitative Results: Table III gives the clustering performance comparison of different methods evaluated on SalinasA, Indian Pines, and Pavia University data sets. As we can see from the results, the proposed GCSC methods achieve the best clustering performance and significantly outperform the other clustering methods in terms of OA, NMI, and Kappa. We can further find the following tendencies from the results.

First, equipped with graph convolution, the traditional subspace clustering models can achieve remarkable improvement compared with the traditional counterpart. For example, EGCSC is significantly better than EDSC. It signifies that the proposed GCSC framework is beneficial for subspace clustering. Furthermore, few of the existing methods we compared with our methods can reach 80% OA, as shown in Table III. On the contrary, OAs yielded by the EGCSC and EKGCSC models are generally better than 84% on all the data sets. Particularly, on the SalinasA data set, EKGCSC achieves a perfect (100%) clustering performance.

Second, EKGCSC outperforms EGCSC on all the three data sets. Due to the complexity of HSI, linear models often cannot fully exploit the relationship among data points. By extending EGCSC into the nonlinear kernel space, EKGCSC's performance can be dramatically enhanced. In other words, EKGCSC considers the nonlinear relationship between data points and makes the learned affinity matrix more robust. In the experiment, EKGCSC achieves 0.15%, 2.78%, and 12.94% improvement on SalinasA, Indian Pines, and Pavia University data sets, respectively.

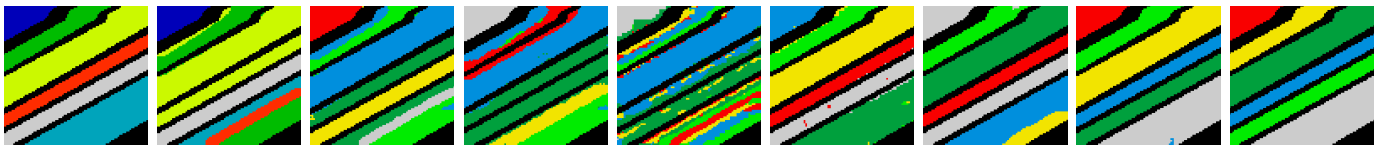
Third, the results obtained by EGCSC and EKGCSC are comparable with many supervised HSI classification methods [5], [9], [55]. Specifically, the EKGCSC model achieves 100%, 87.61%, and 97.36% in terms of OA on the SalinasA, Indian Pines, and Pavia University data sets, respectively. The recent development of supervised HSI classification allows achieving excellent results. However, the unsupervised classification of HSI is still a challenging task. The state-of-the-art clustering

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TABLE III

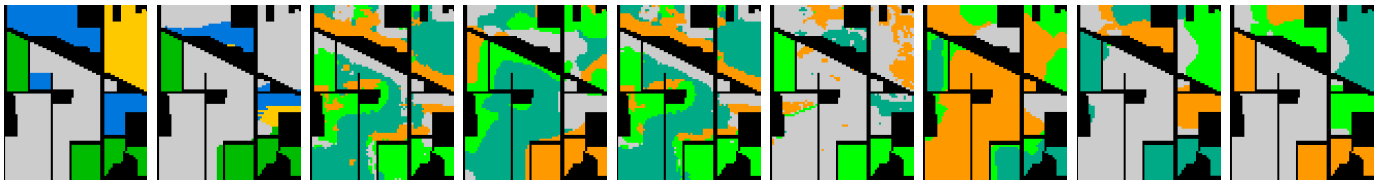
CLUSTERING PERFORMANCE OF THE COMPARED METHODS ON INDIAN PINES, SALINASA, AND PAVIAU DATA SETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Data	Metric	SC [24]	SSC [27]	LRSC [36]	ℓ_2-SSC [22]	S^{4}C [23]	UBL [49]	RMMF [18]	EDSC [51]	EGCSC	EKGCSC
SaA.	OA	0.6806	0.7666	0.5613	0.6412	0.8631	0.9142	0.9820	0.8702	0.9985	**1.0000**
	NMI	0.7464	0.7571	0.4242	0.6971	0.7977	0.8692	0.9483	0.9135	0.9949	**1.0000**
	Kappa	0.6002	0.7138	0.4487	0.5546	0.8312	0.8943	0.9775	0.8384	0.9981	**1.0000**
InP.	OA	0.6841	0.4937	0.5142	0.6645	0.7008	0.6258	0.7121	0.7126	0.8483	**0.8761**
	NMI	0.5339	0.2261	0.2455	0.3380	0.5445	0.6680	0.4985	0.4717	0.6422	**0.6959**
	Kappa	0.5055	0.2913	0.3145	0.5260	0.5825	0.4690	0.5609	0.5657	0.6422	**0.8211**
PaU.	OA	0.7691	0.6146	0.4326	0.5842	0.6509	0.7083	0.7704	0.6175	0.8442	**0.9736**
	NMI	0.6784	0.6545	0.3793	0.4942	0.7031	0.6874	0.7388	0.5750	0.8401	**0.9529**
	Kappa	0.8086	0.4886	0.2549	0.3687	0.5852	0.6533	0.6804	0.4250	0.7968	**0.9653**



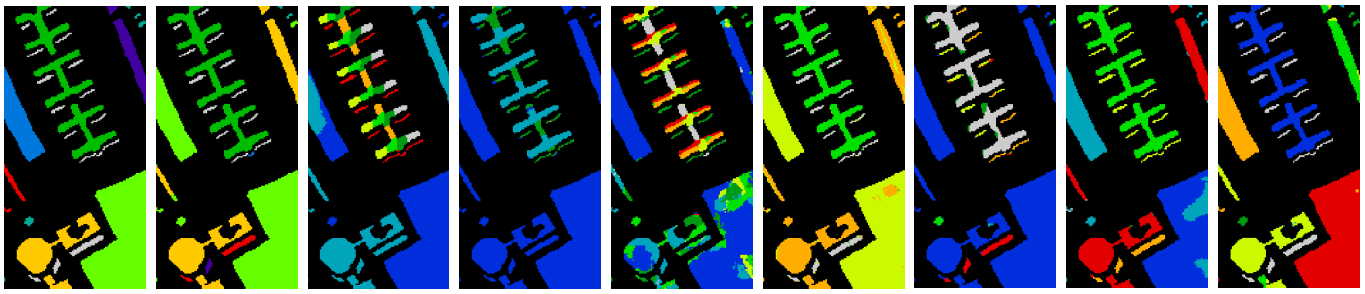
(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 4. Clustering results obtained by different methods on the SalinasA data set. (a) Ground truth. (b) SC 68.06%. (c) SSC 76.66%. (d) ℓ_2-SSC 64.12%. (e) LRSC 56.13%. (f) RMMF 98.20%. (g) EDSC 87.02%. (h) EGCSC 99.85%. (i) EKGCSC 100%.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 5. Clustering results obtained by different methods on the Indian Pines data set. (a) Ground truth. (b) SC 68.41%. (c) SSC 49.37%. (d) ℓ_2-SSC 66.45%. (e) LRSC 51.42%. (f) RMMF 71.20%. (g) EDSC 71.26%. (h) EGCSC 84.83%. (i) EKGCSC 87.61%.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 6. Clustering results obtained by different methods on the Pavia University data set. (a) Ground truth. (b) SC 76.91%. (c) SSC 64.46%. (d) ℓ_2-SSC 58.42%. (e) LRSC 43.26%. (f) RMMF 77.04%. (g) EDSC 61.75%. (h) EGCSC 84.42%. (i) EKGCSC 97.36%.

performance of our methods bridges the gap between unsupervised HSI classification and supervised HSI classification.

2) Results Visualization: To visually observe the clustering results, we visualize the clustering maps of different clustering methods in Figs. 4–6. Since the source codes of S^{4}C and UBL have not been released, their class maps are not included in the figures. Nevertheless, it does not affect the analysis. Notice that the color of the same class may change in different class maps. This is because label values can be permuted by different clustering methods. Fig. 4 shows that the class map obtained by EKGCSC on the SalinasA data set is in complete agreement with the ground truth. For the Indian Pines and Pavia University data sets (i.e., Figs. 5 and 6), EKGCSC shows the best class maps that are closest to the ground truths. Compared with the other competitors, EGCSC shows better class maps. While the class maps obtained by the other methods (e.g., SSC, LRSC, and EDSC) contain relatively more noisy points caused by misclassification. Briefly, the results

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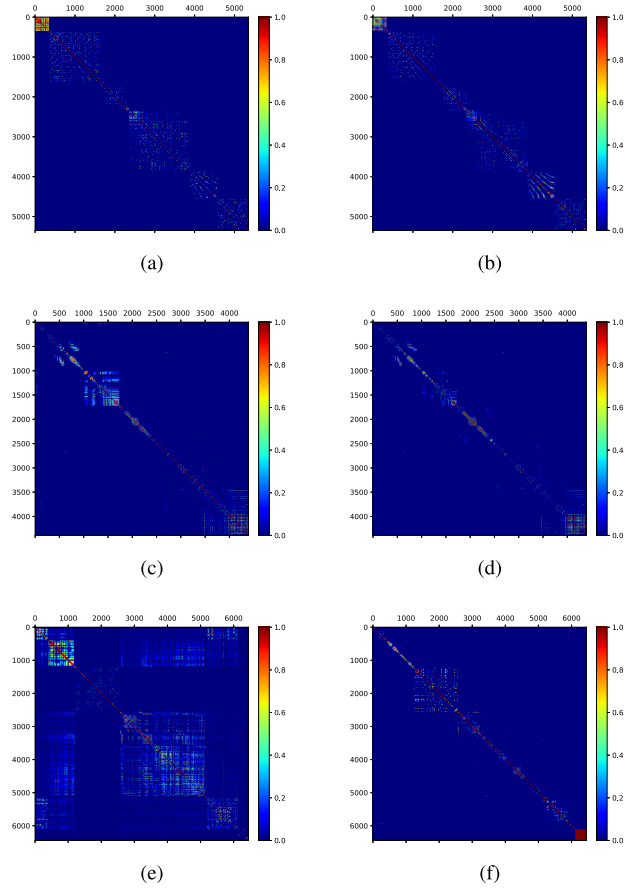


Fig. 7. Visualization of the obtained affinity matrix of EGCSC and EKGCSC on (a) and (b) SalinasA, (c) and (d) Indian Pines, and (e) and (f) Pavia University data sets.

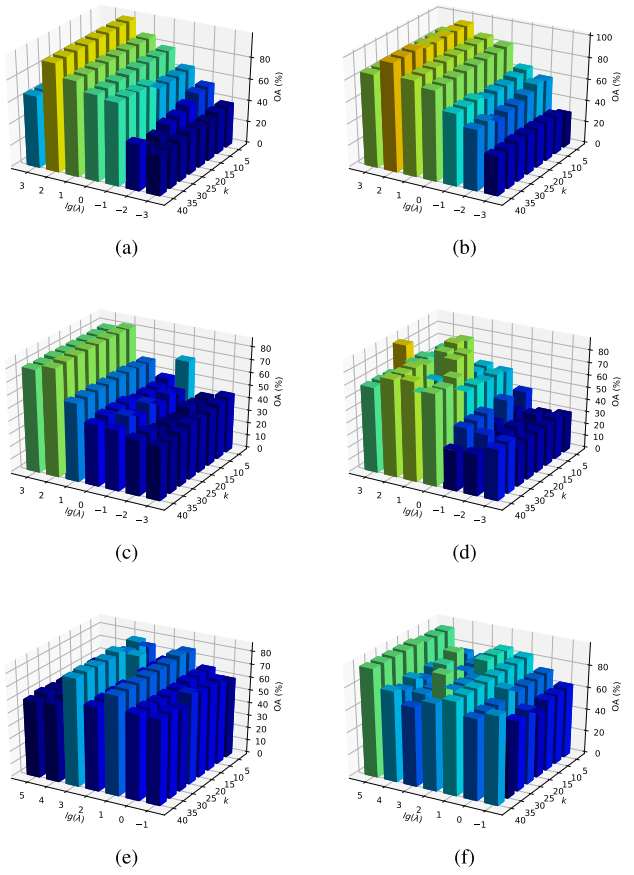


Fig. 8. Influence of λ and k for EGCSC and EKGCSC on (a) and (b) SalinasA, (c) and (d) Indian Pines, and (e) and (f) Pavia University data sets.

demonstrate the effectiveness and superiority of the proposed GCSC framework.

3) Visualization of the Learned Affinity Matrix: In Fig. 7, we visualize the affinity matrices learned by the EGCSC and EKGCSC models. For a better presentation, we have reordered data points according to the ground truth before computing the affinity matrix. In the figures, each column or row of the affinity matrix denotes the self-representation coefficients that using all data points to represent the corresponding data point. Therefore, the larger the coefficient is, the more the corresponding data point contributes to the reconstruction. Ideally, if a group of data points belong to the same cluster, their self-representation coefficients to each other will be nonzero. Otherwise, they will be zero. Thus, an ideal affinity matrix is block-diagonal. From Fig. 7(a)–(f), we can observe that the obtained affinity matrices by both EGCSC and EKGCSC are sparse and have an apparent block-diagonal structure. Furthermore, EKGCSC shows better block-structure than EGCSC, which demonstrates that EKGCSC can more accurately explore the intrinsic relationships between data points and thus achieve better performance.

4) Impact of λ and k: In this experiment, we investigate the impact of the two most important hyperparameters involved in the GCSC framework, i.e., the regularization coefficient λ and the number of the nearest neighbors k for the kNN graph. We set λ in the range of $[10^{-3}, 10^{3}]$ for SalinasA and Indian Pines data sets, and $[10^{-1}, 10^{5}]$ for Pavia University data set. For clarity, we take $lg(\lambda)$ into account for plotting. For all data sets, we let k vary from 5 to 40 with an interval of 5. The results are shown in Fig. 8. It can be seen that λ has a significant impact on clustering performance. We can further observe a tendency, i.e., the clustering performance will increase as λ increased. By contrast, both EGCSC and EKGCSC are insensitive to k. However, when k is too large, graph convolution will result in an oversmoothing problem. That is, the graph embedding of all the data points will become similar. Therefore, too large k may negatively affect the clustering performance. According to the empirical study, we provide a group of the best hyperparameter setting in Table II. In practice, the selection of parameters often varies with data sets. An effective way to do so is to use a grid search. Empirically, the suitable λ and k can be found in $[10, 10^{5}]$ and $[5, 30]$, respectively.

5) Impact of the Number of PCs: Although dimensionality reduction is an optional step in our framework, it often provides a good balance between computational efficiency and model performance. To empirically study the influence of the number of PCs, we perform the proposed methods with varying PCs from 1 to 8. We show the corresponding OA and CPV in Fig. 9. As shown in Fig. 9(a) and (b), the

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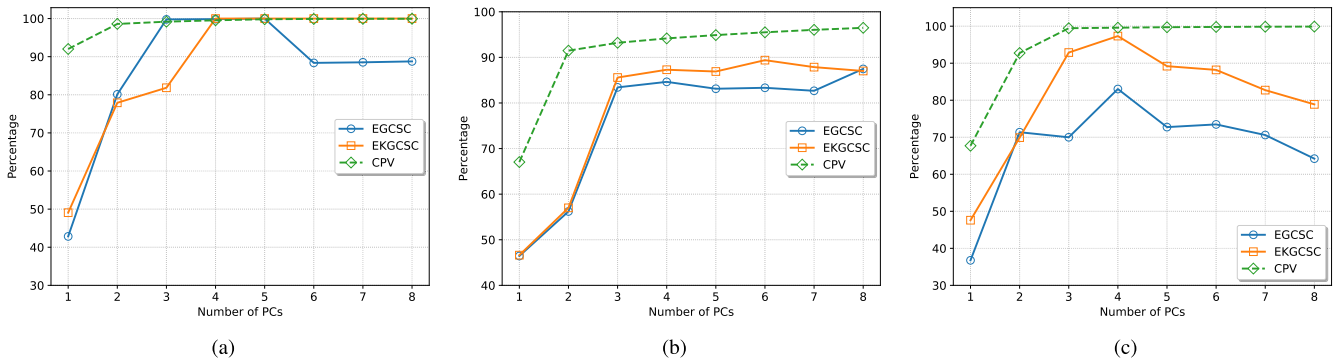


Fig. 9. Clustering OA under varying number of PCs for (a) SalinasA, (b) Indian Pines, and (c) Pavia University data sets.

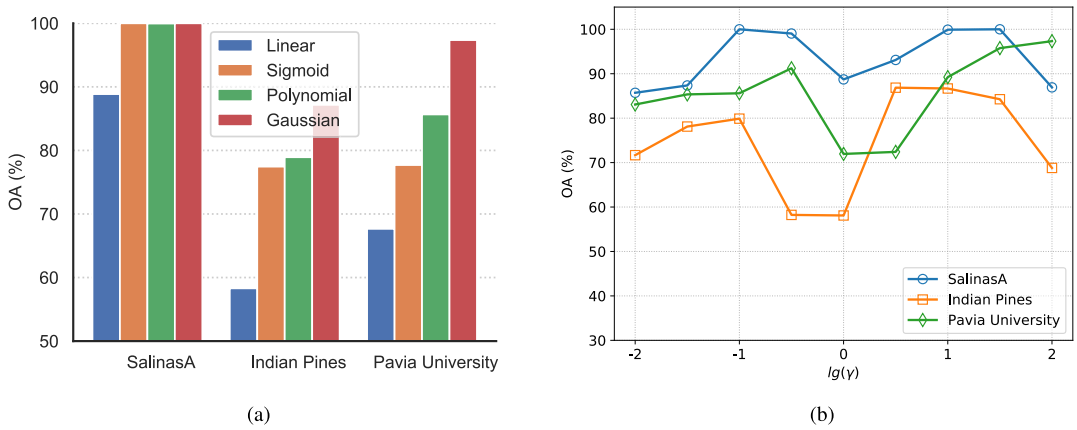


Fig. 10. Comparison of different kernel functions and different parameter settings in EKGCSC. (a) Comparison of different kernel functions. (b) Comparison of different parameter settings of Gaussian kernel function.

clustering OA of the GCSC models improves with PCs. This is given by the fact that more discriminative information is included when more PCs are considered. However, it is not always the case that more PCs will lead to better clustering performance. For example, in Fig. 9(c), the clustering OA increases first and then decreases. The phenomenon is known as the *Hughes* phenomenon [56], which mainly results from the increased correlation and insufficient data points. As we can observe from Fig. 9(c), the top 3 PCs of Pavia University data set provide 99.45% CPV. This signifies that the remainder PCs contain very little information, hence increasing the risk of introducing redundancy. Therefore, to correctly detect subspaces, the model has to use more data points to model the statistical property of data. Nevertheless, EKGCSC shows significantly better OA than EGCSC under different PCs.

6) Impact of Kernel Functions: Kernel function plays an important role in EKGCSC. In this experiment, we explore the impact of kernel functions. First, we compare four widely-used kernel functions, i.e., linear kernel, sigmoid kernel, polynomial kernel, and Gaussian kernel, on the three HSI data sets. The results are shown in Fig. 10(a). Among the four results, the Gaussian kernel achieves the best clustering performance on all three data sets, while the linear kernel is significantly lower than the other compared kernels. Compared with the sigmoid kernel, the polynomial kernel shows better OA on Indian Pines and Pavia University data set. In summary, the results suggest that the Gaussian kernel is the more suitable kernel function for the proposed EKGCSC model. Second, we analyze the influence of the parameter of the Gaussian kernel, i.e., γ. We show the results in Fig. 10(b), where γ varies in the range from 0.01 to 100. It can be seen that, for all data sets, the performance curve shows a similar tendency for different data sets and the best γ roughly locates within two regions, i.e., $[0.1, 1]$ and $[5, 50]$. Notice that, when $\gamma = 1$, EKGCSC generally obtain worse clustering performance on all three data sets. A group of suitable settings of γ used in our experiment is given in Table II.

7) Comparison of Running Time: We compare our methods with the other competitors in terms of running time. Table IV lists the running time of different clustering methods. Since EGCSC and EKGCSC have closed-form solutions without needing an iterative operation, they are significantly faster than SSC, LRSC, S^4C, and UBL. Although SC, ℓ_2-SSC, and RMMF take less running time, they cannot achieve a better performance than our methods. Compared to EDSC, our methods take relatively more running time, which is because the proposed GCSC framework needs to construct the graph from data points. Furthermore, EKGCSC needs to compute the kernel matrix, thus its running time will be increased compared with EGCSC. To sum up, our proposed EGCSC

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TABLE IV
RUNNING TIME OF DIFFERENT METHODS (IN SECOND)

Data	SC [24]	SSC [27]	LRSC [36]	ℓ_2-SSC [22]	S^4C [23]	UBL [49]	RMMF [18]	EDSC [51]	EGCSC	EKGCSC
SaA.	13.203	855.663	7030.710	4.717	9363.5	2509.68	9.448	42.762	92.951	131.485
InP.	8.587	653.998	3980.004	3.272	1567.9	104.90	1.494	24.093	69.366	99.052
PaU.	15.640	1022.382	15861.621	15.677	7398.3	237.44	4.310	98.533	124.333	264.649

and EKGCSC models achieve a good balance between time cost and clustering accuracy.

V. CONCLUSION AND FUTURE WORKS

We proposed a novel HSI clustering framework, termed as GCSC, based on introducing graph convolution into subspace clustering. The key to the proposed framework is to utilize a graph convolutional self-representation to incorporate the intrinsic structure information of data points. Traditional subspace clustering models can be treated as the special forms of the GCSC framework built on the Euclidean data. Benefiting from the graph convolution, the GCSC model tends to use a clear dictionary to learn a robust affinity matrix. We designed two efficient subspace clustering models (i.e., EGCSC and EKGCSC) based on the proposed GCSC framework by using the Frobenius norm. The experimental results on three HSI data sets demonstrate that the proposed GCSC models can achieve state-of-the-art performance with significant margins compared with many existing clustering models. Particularly, the EKGCSC model achieves 100%, 87.61%, and 97.36% clustering OA on SalinasA, Indian Pines, and Pavia University data sets, respectively.

The successful attempt of the GCSC model signifies that considering the intrinsic graph structure among data sets is important for clustering. It offers an alternative orientation for unsupervised learning. The proposed GCSC framework also enables us to revisit traditional clustering models in the non-Euclidean domain. This means that any improved strategies used in the classical subspace clustering models are applicable to GCSC. Moreover, it should be noticed that the proposed GCSC framework suffers from quadratic complexity in the size of data points, making it hard to be used for large-scale HSI data. The potential solutions to this issue will be studied in our future work.

APPENDIX

Proof: The solution of EGCSC.

Let $\mathcal{L}$ be the loss function of EGCSC and (7) can be rewritten as

$$\begin{aligned}\mathcal{L}(\mathbf{Z}) &= \frac{1}{2}\left\|\mathbf{X}\bar{\mathbf{A}}\mathbf{Z}-\mathbf{X}\right\|_F^2+\frac{\lambda}{2}\|\mathbf{Z}\|_F^2\\ &= \frac{1}{2}\mathrm{tr}\Big[\left(\mathbf{X}\bar{\mathbf{A}}\mathbf{Z}-\mathbf{X}\right)^T\left(\mathbf{X}\bar{\mathbf{A}}\mathbf{Z}-\mathbf{X}\right)+\lambda\mathbf{Z}^T\mathbf{Z}\Big]\\ &= \frac{1}{2}\mathrm{tr}\big(\mathbf{Z}^T\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}}\mathbf{Z}+\mathbf{X}^T\mathbf{X}-2\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}}\mathbf{Z}+\lambda\mathbf{Z}^T\mathbf{Z}\big).\end{aligned} \tag{15}$$

According to the properties of matrix trace and matrix derivatives, the partial derivative of $\mathcal{L}$ with respect to $\mathbf{Z}$ can be presented as

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial\mathbf{Z}} &= \bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}}\mathbf{Z}-\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}+\lambda\mathbf{Z}\\ &= \big(\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}}+\lambda\mathbf{I}_N\big)\mathbf{Z}-\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}.\end{aligned} \tag{16}$$

Let $(\partial\mathcal{L}/\partial\mathbf{Z})=0$, we get

$$\big(\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}}+\lambda\mathbf{I}_N\big)\mathbf{Z}=\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}. \tag{17}$$

Finally, $\mathbf{Z}$ can be expressed as

$$\mathbf{Z}=\big(\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}}+\lambda\mathbf{I}_N\big)^{-1}\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}. \tag{18}$$

Due to $\bar{\mathbf{A}}^T\mathbf{X}^T\mathbf{X}\bar{\mathbf{A}}+\lambda\mathbf{I}$ is positive semidefinite, its reversible always exists. ■

Proof: The solution of EKGCSC.

Similar to the above proof, the loss function of EKGCSC [see (10)] can be expressed as

$$\begin{aligned}\mathcal{L}(\mathbf{Z}) = \frac{1}{2}\mathrm{tr}\big(&\mathbf{Z}^T\bar{\mathbf{A}}^T\mathbf{K}_{\mathbf{XX}}\bar{\mathbf{A}}\mathbf{Z}-2\mathbf{K}_{\mathbf{XX}}\bar{\mathbf{A}}\mathbf{Z}\\ &+\mathbf{K}_{\mathbf{XX}}+\lambda\mathbf{Z}^T\mathbf{Z}\big).\end{aligned} \tag{19}$$

The partial derivative of $\mathcal{L}$ with respect to $\mathbf{Z}$ is then given by

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial\mathbf{Z}} &= \bar{\mathbf{A}}^T\mathbf{K}_{\mathbf{XX}}\bar{\mathbf{A}}\mathbf{Z}-\bar{\mathbf{A}}^T\mathbf{K}_{\mathbf{XX}}+\lambda\mathbf{Z}\\ &= \big(\bar{\mathbf{A}}^T\mathbf{K}_{\mathbf{XX}}\bar{\mathbf{A}}\mathbf{Z}+\lambda\mathbf{I}_N\big)\mathbf{Z}-\bar{\mathbf{A}}^T\mathbf{K}_{\mathbf{XX}}.\end{aligned} \tag{20}$$

By setting $(\partial\mathcal{L}/\partial\mathbf{Z})=0$, we finally get the optimal solution of $\mathbf{Z}$ as follows:

$$\mathbf{Z}=\big(\bar{\mathbf{A}}^T\mathbf{K}_{\mathbf{XX}}\bar{\mathbf{A}}+\lambda\mathbf{I}_N\big)^{-1}\bar{\mathbf{A}}^T\mathbf{K}_{\mathbf{XX}}. \tag{21}$$

■

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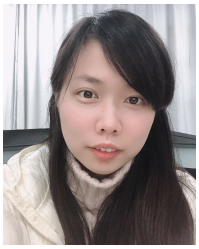
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